

PRATYUSH PADHY

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EDUCATION

University of California, Irvine

Irvine, CA

B.S., Computer Science, Specialization in Intelligent Systems, GPA: 3.86/4.00

Expected Graduation: 2028

- **Dean's Honor List:** Fall '25, Winter '26, Spring '26 | **Organizations:** Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS)

East Palo Alto, CA

Incoming Software Development Engineer Intern

Sept. 2026 – Dec. 2026

- Open Data Analytics Engines (ODAE) Runtime: distributed query execution for Spark, Trino, and Flink on EMR and Athena.

Tata Consultancy Services (TCS)

CMU Campus, Pittsburgh, PA

AI/ML Intern, TCS Medical Robotics Center

Jun. 2026 – Aug. 2026

- Benchmarked task-specific RL against a generalist vision-language-action (VLA) policy (SmolVLA, LeRobot) over **250+ episodes** on an AgileX Piper 6-DOF arm; the VLA matched state-based RL (**85.3% vs. 81.5%** for PPO) from pixels alone.
- Diagnosed a sparse-reward exploration collapse in simulation and fixed it with action-space reparameterization (no reward shaping or demonstrations), training PPO (Stable-Baselines3) to **81.5% $\pm$ 3.3%** liquid-transfer success across **3 independent seeds**.
- Budget-matched PPO on identical pixels scored **0%** where SmolVLA, fine-tuned in **13 min** via two-stage imitation learning (real-robot pretraining, then 120 PPO-generated demos), succeeded; an ablation isolating pretrained vision + imitation.
- Extended FluidLab (ICLR '23 MPM fluid simulator) into a robotics platform for liquid manipulation: URDF forward-kinematics joint control, analytic SDF liquid vessels on a 128^3 voxel grid, an offscreen renderer, and **3 upstream engine-bug fixes**.

Ready Tutor

Irvine, CA

Software Developer Intern

Mar. 2026 – Jun. 2026

- Designed a centralized storage layer: a storage-agnostic schema class over MongoDB/SQLite behind a shared React useContext provider, replacing **10+** scattered storage instances and batching persistence into **a single write per user action**.
- Built a course catalog with real-time retrieval of **1,200+ UCI courses**, the headline feature of the funding demo that secured **\$5,500**; resolved **30+ production bugs** across the stack, including a credits-calculation error affecting schedule planning.
- Redesigned the year planner from a rigid 4-year structure to support unlimited years with per-quarter course blocks.

PROJECTS

Chronicle | Python, FastAPI, Postgres/pgvector, Next.js | chronicles-weld.vercel.app

Jun. 2026 – Sept. 2026

- Built a job aggregation platform ingesting **~26K open roles** from **600+ companies'** ATS boards (Greenhouse, Lever, Ashby) via async fan-out with per-company fault isolation and idempotent 24–48h refreshes that dedupe cross-source postings.
- Engineered hybrid search fusing GIN-indexed Postgres full-text with semantic ranking (int8 ONNX MiniLM, pgvector HNSW) via RRF at **p95 19ms**; a profile-vector recommender lifted NDCG@10 **0.58→0.81** over the rule baseline (n=24 personas).
- Shipped Google OAuth accounts, kanban application tracking, saved-search email alerts, scheduled daily ingestion (GitHub Actions cron with an idempotent run-lock), a **119-test CI suite** gating every push, and a keep-warm probe against cold starts.

Forge | Python, PyTorch, Unsloth, TRL, vLLM, Ollama | forge-grpo.vercel.app

Jul. 2026 – Sept. 2026

- Post-trained Qwen2.5-1.5B to reason via GRPO (RL from verifiable rewards, no human labels or reward model) with Unsloth + TRL, lifting GSM8K pass@1 **58.8%→70.0%** on 1,319 held-out problems in **86 min** at **3.64 GiB peak VRAM**.
- Designed the reward stack (correctness, format, graded tag-presence), diagnosing a zero-reward cold start that nulled GRPO's gradient; verified **100% output-format compliance** and no catastrophic forgetting on ARC-Challenge (69.5%→68.5%).
- Quantized to GGUF Q4_K_M (**3.15× smaller**) and served at **228 tok/s** (Ollama) and **674 tok/s** batched (vLLM multi-LoRA) behind a live base-vs-tuned playground running real streaming inference on a scale-to-zero GPU endpoint.

FacePulse | Python, TensorFlow, FastAPI, React, OpenCV, dlib | face-pulse.netlify.app

Apr. 2026 – Jun. 2026

- **Featured on the Handshake AI Showcase**, selected from thousands of student submissions; evolved from a facial-landmark detector into a comparative study of 4 architectures (pixel MLP, landmark MLP, custom CNN, two-phase VGG16) on FER2013 (35K+ images, 5 classes), the custom CNN leading at **64.5% accuracy / 0.647 macro-F1**, ahead of all three baselines.
- Built a full-stack app: live in-browser webcam detection (React + face-api.js) with real-time emotion probability charts and session analytics (Chart.js), plus a FastAPI + TensorFlow inference API serving the trained checkpoints (Docker, CI).

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML/CSS, Bash

ML & Concepts: reinforcement learning (PPO, GRPO), imitation learning & VLA fine-tuning, RAG & hybrid retrieval, LLM fine-tuning & quantization, embeddings & vector search, recommender systems, evaluation & ablations, NLP

Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, Transformers, Unsloth, TRL, vLLM, LeRobot, Stable-Baselines3, Gymnasium, OpenCV, NumPy, Pandas, Matplotlib, React, Next.js, FastAPI, SQLAlchemy, Tailwind CSS

Tools & Platforms: Git, GitHub Actions (CI), Docker, pytest, PostgreSQL, Neon, Ollama, ROS2, Gazebo, FluidLab, WSL2/CUDA, Vercel, Render, MongoDB, SQLite, Modal, Netlify